

Introduction to Data Science

H. M. Samadhi Chathuranga Rathnayake
MSc, MBA, PGDip, BSc (1st Class)

Why Are We Talking About Data Today?

- Every action we take creates data
- Mobile apps, social media, banks, hospitals, businesses all depend on data
- Decisions today are no longer based on intuition alone
- Organizations that use data effectively gain strong competitive advantages



What Is Data?

- Data is a collection of raw facts and observations
- Data by itself has no meaning until it is processed
- Examples: numbers, text, images, videos, sensor readings
- Data becomes valuable only when converted into information and insights



From Data to Knowledge



- **Data** → raw facts
- **Information** → processed and organized data
- **Knowledge** → understanding patterns and relationships
- **Wisdom** → making decisions based on knowledge
- Data Science focuses on moving up this ladder effectively

What Is Data Science?

- Data Science is an interdisciplinary field
- It combines:
 - Statistics
 - Computer Science
 - Domain Knowledge
- The goal is to extract useful insights from data for decision-making

Simple Definition of Data Science

- Data Science is the practice of:
 - Collecting data
 - Cleaning data
 - Analyzing data
 - Interpreting results



Outcomes of Data Science

- It helps answer questions like:

- What happened?

- Why did it happen?

- What will happen next?

- How to optimize the processes based on insights?

Descriptive Analytics

Inferential Analytics

Predictive Analytics

Prescriptive Analytics



Why Data Science Is Important

Helps organizations make evidence-based decisions

Reduces uncertainty and risk

Improves efficiency and performance

Enables prediction and automation

Drives innovation across industries

Real-World Applications of Data Science



Healthcare: disease prediction, patient monitoring



Finance: fraud detection, credit scoring



Marketing: customer segmentation, personalization



Education: student performance analysis



Transportation: traffic prediction, route optimization

Data Science vs Traditional Decision Making

Traditional
decisions

- Based on experience and intuition

Data-driven
decisions

- Based on evidence and analysis

Data Science
supports scalable
and repeatable
decision
processes

Types of Data

Structured Data

- Tables, databases, spreadsheets

Semi-Structured Data

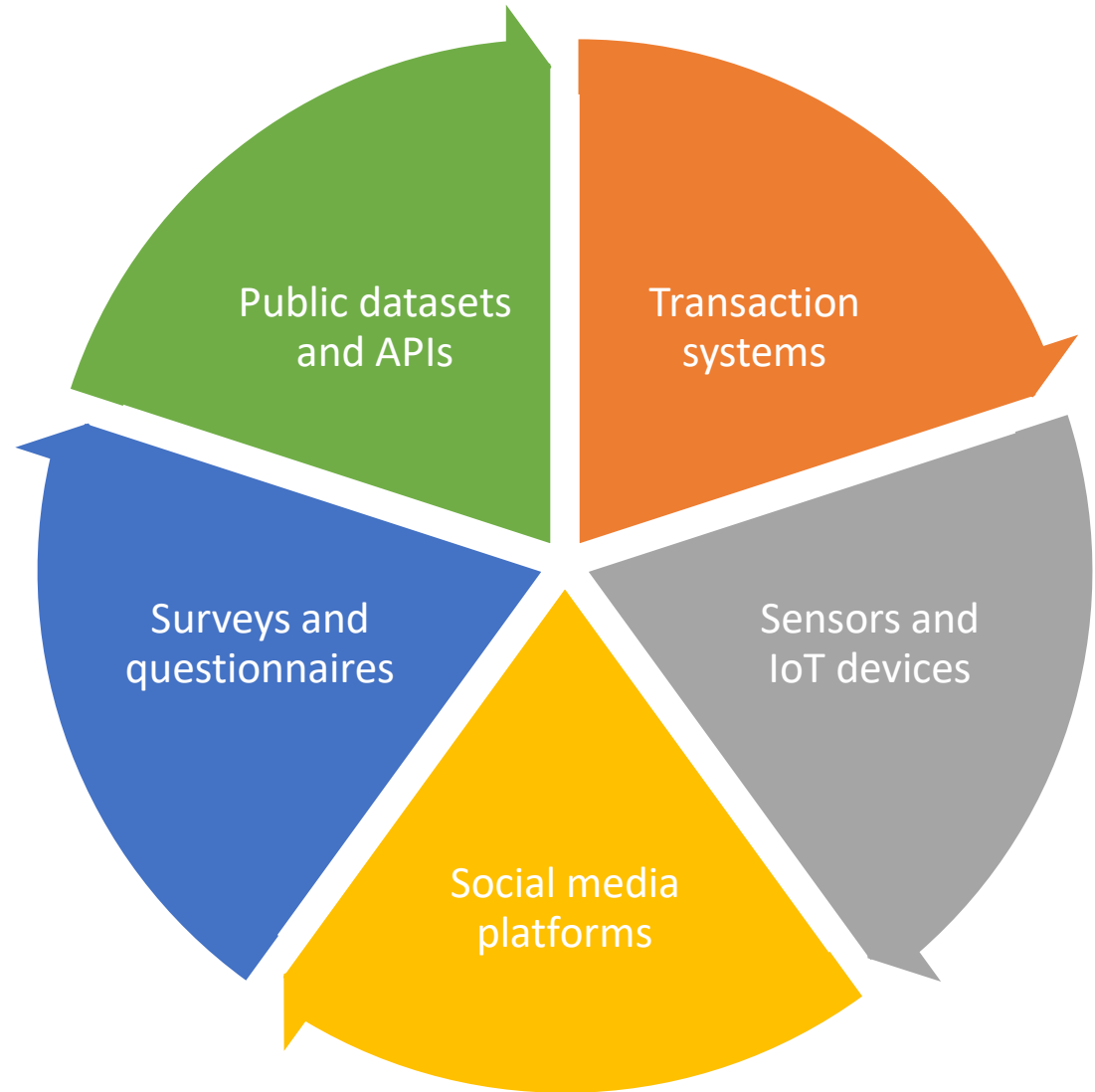
- JSON, XML, logs

Unstructured Data

- Text, images, audio, video

Majority of real-world data is unstructured

Sources of Data





The Data Science Ecosystem

Data storage
systems

Data
processing
tools

Programming
languages

Visualization
tools

Machine
learning
frameworks

Cloud
platforms

Pipeline of Data Science

Step 1 - Data Collection

Step 2 - Data Cleaning and Preparation

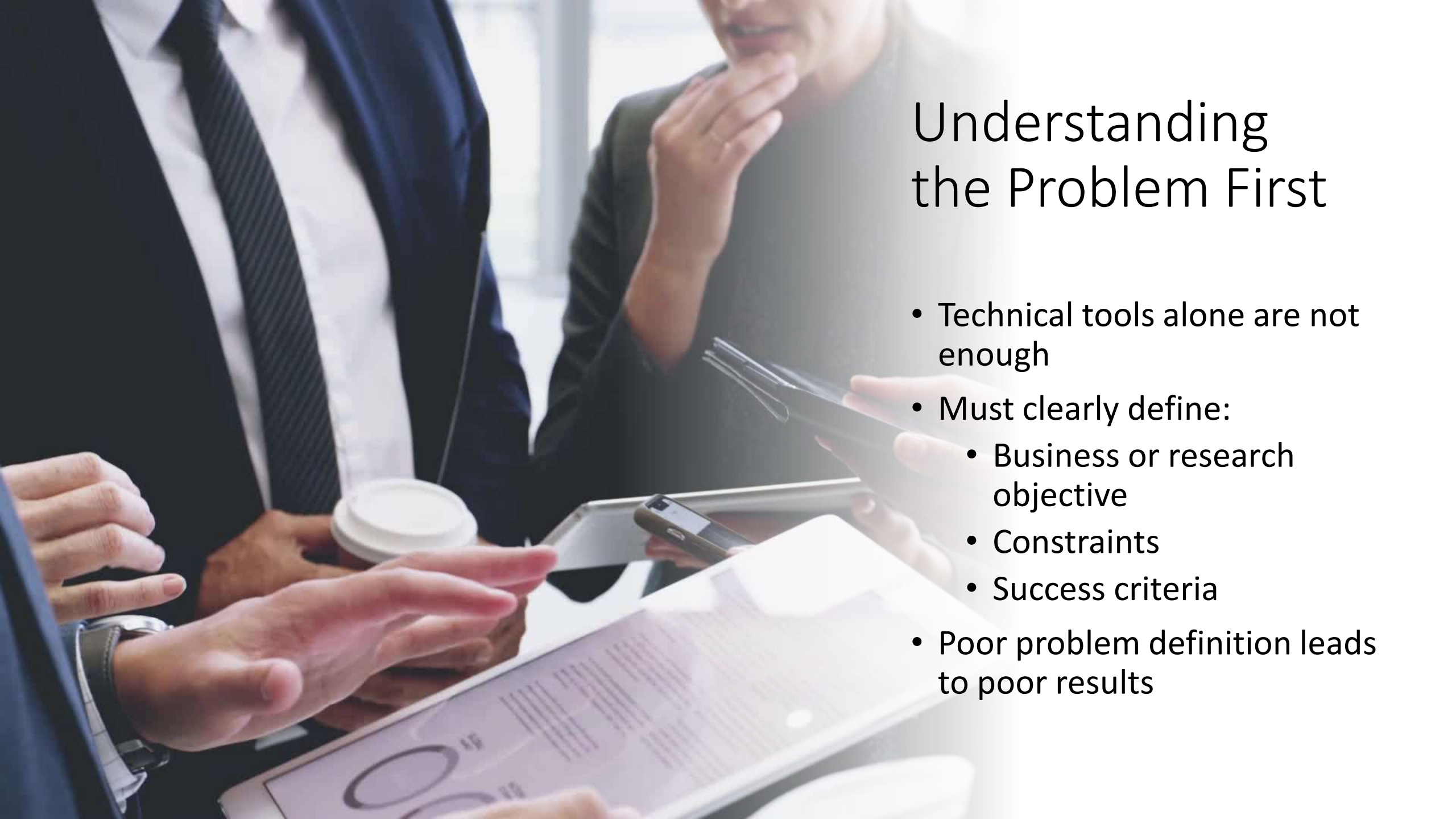
Step 3 - Exploratory Data Analysis

Step 4 - Modeling and Algorithms

Step 5 - Evaluation and Interpretation

Step 6 - Communication of results (Data storytelling)

Additional Step – Deploying the Systems

A blurred background image of a business meeting. Several people in professional attire are gathered around a table. One person is holding a tablet displaying a chart, while others are gesturing or holding coffee cups. The scene is brightly lit, suggesting an office environment with large windows.

Understanding the Problem First

- Technical tools alone are not enough
- Must clearly define:
 - Business or research objective
 - Constraints
 - Success criteria
- Poor problem definition leads to poor results

Data Collection and Data Quality

Data quality directly affects results

Common data issues:

- Missing values
- Noise
- Duplicates
- Inconsistent formats

Cleaning is often the most time-consuming step

Exploratory Data Analysis (EDA)

EDA helps understand data before modeling

Uses:

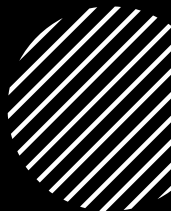
- Summary statistics
- Charts and graphs

Identifies:

- PatternsTrends
- Outliers



Role of Statistics in Data Science



Statistics is the foundation of the Data Science



Describes data behavior



Helps quantify uncertainty



Enables hypothesis testing



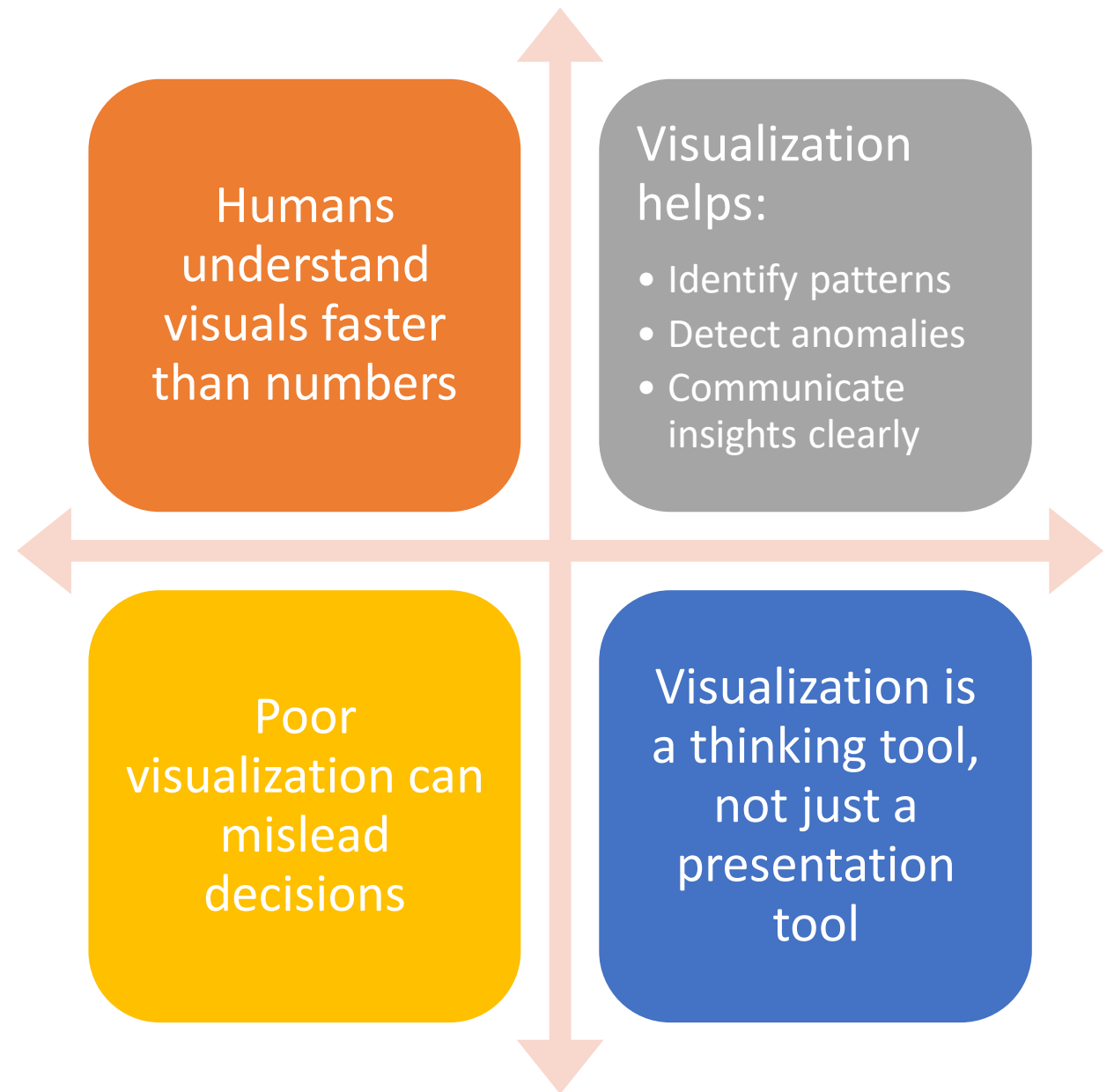
Supports inference and prediction



Forms the foundation of analytical thinking



Role of Visualization in Data Science



Role of Programming in Data Science

Automates data processing

Handles large datasets efficiently

Enables reproducible analysis

Common languages:

- Python
- R

Programming turns ideas into working solutions

Role of Machine Learning

Allows systems to learn from data

Used for:

- Prediction
- Classification
- Clustering

Machine learning is a subset of Data Science and Artificial Intelligence

Data Science vs Machine Learning

Data Science:

- End-to-end process
- Includes data handling, analysis, and communication

Machine Learning:

- Focuses on building predictive models

Not all Data Science uses Machine Learning



Data Scientist: Who Are They?

- A Data Scientist is a problem solver
 - Combines:
 - Analytical thinking
 - Technical skills
 - Communication skills
 - Acts as a bridge between data and decision makers
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Skills Required for Data Science



Mathematical and statistical reasoning

Programming and computational thinking

Data visualization

Domain understanding

Critical thinking and curiosity



Data Science Mindset

- Curiosity-driven thinking
- Skeptical attitude towards results
- Willingness to question assumptions
- Comfort with uncertainty
- Continuous learning and adaptation



Ethical Considerations in Data Science

- Data privacy and security
 - Bias and fairness
 - Responsible use of data
 - Transparency in models
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- Ethical decisions are as important as technical accuracy



Challenges in Data Science



Poor data
quality

Data privacy
concerns

High
dimensional
data

Interpretability
of models

Changing data
over time